# PHYS 7502: Particles and Nuclei 2

Final Assignment — Complete Study Guide

Hadrons, Hadronization, Jets, String Model, DPM/Pomeron, Bag Model

Instructor: Prof. Justin Frantz

Spring 2022 / Study Guide prepared May 2026

## Contents

**1 Part I: Jets and Event Generators (Questions 1–7) 2**
- 1.1 Question 1: Most Common Event Generator . . . 2
- 1.2 Question 2: Hadronization Model in PYTHIA . . . 3
- 1.3 Question 3: Two Main Kinds of Jet Finding Algorithms . . . 4
- 1.4 Question 4: Which Algorithm Is No Longer Used? . . . 5
- 1.5 Question 5: Jet Algorithm Used at LHC . . . 5
- 1.6 Question 6: The Variable $R$ and Its Formula . . . 6
- 1.7 Question 7: Units of $R$ and Why . . . 7

**2 Part II: Fragmentation Functions (Questions 8–10) 9**
- 2.1 Question 8: Argument of the Fragmentation Function $D()$ . . . 9
- 2.2 Question 9: Three Quantities in Factorization Convolution . . . 10
- 2.3 Question 10: Quark vs. Gluon Jets — DELPHI Results . . . 11

**3 Part III: Schwinger Mechanism and String Model (Questions 11–18) 12**
- 3.1 Optional Warmup: Eq. 5.2 and E&M . . . 12
- 3.2 Question 11: Order of Magnitude of String Tension $\kappa$ . . . 12
- 3.3 Question 12: Difference Between Eq. 5.4 and 5.5 . . . 13
- 3.4 Question 13: Diagram for Each Value of WHAT? . . . 14
- 3.5 Question 14: Regge Trajectory . . . 14
- 3.6 Question 15: Experimental Data the String Model Cannot Explain . . . 15
- 3.7 Question 16: Figure 7.4(a) — Slopes and Rest/Motion . . . 15
- 3.8 Question 17: The Yo-Yo Model . . . 16
- 3.9 Question 18: Geometric Curve for Particle Production Points (Lund Model) . . . 17

**4 Part IV: Dual Parton Model / Pomeron (Questions 19–23) 18**
- 4.1 Question 19: Hadron Produced in Diffractive Events . . . 18
- 4.2 Question 20: Color Dipole Model . . . 18
- 4.3 Question 21: Veneziano Amplitude Symmetry . . . 18
- 4.4 Question 22: Total Cross Section vs. CM Energy . . . 19
- 4.5 Question 23: Pomeron Quantum Numbers . . . 19

**5 Part V: MIT Bag Model and QGP (Questions 24–28) 20**
5.1 Question 24: Important Variables in the Bag Model . . . . . 20
5.2 Question 25: What Does $B$ Represent? . . . . . 20
5.3 Question 26: How the 1/4 Power Arises . . . . . 21
5.4 Question 27: Two Physical Quantities Derived from the Bag Model . . . . . 22
5.5 Question 28: Critical Temperature for QGP Formation . . . . . 22

**6 Summary Table 25**

# 1 Part I: Jets and Event Generators (Questions 1–7)

## 1.1 Question 1: Most Common Event Generator

**Question 1**

What is the most common event generator used to describe tree-level (Leading Order) QCD processes and jets?

**Deliverable**

A single name: the most widely-used LO event generator for QCD jets.

**Background: What is an Event Generator?**

In high-energy particle physics, we cannot solve QCD analytically for the full process of producing jets. Instead, we rely on **Monte Carlo event generators**—computer programs that:

1. Calculate the **tree-level (Leading Order, LO)** cross sections of parton–parton scattering using perturbative QCD Feynman diagrams.
2. Use "global fit" **Parton Distribution Functions (PDFs)** to describe the structure of incoming hadrons.
3. Apply tuned **empirical hadronization models** to convert the outgoing partons into observable hadrons.

The term "tree-level" means the lowest-order Feynman diagrams—no loops. These are also called "Leading Order" (LO) calculations. The simulation uses stochastic (random number) sampling, hence "Monte Carlo."

**Source:** Frantz reading assignment, slides on "High Energy Event Generators" (p. 5 of reading PDF).

**Strategy**

- The reading slides list two LO generators: **PYTHIA** and **HERWIG**.
- The slides state that PYTHIA "became the most widely used by about 2005" and is described as "Most successful."
- HERWIG is described as "No longer used."

**Answer**

**PYTHIA** (the PYTHIA computer program/package, available at `https://pythia.org/`). PYTHIA became the dominant event generator by ∼2005, five years before the LHC started. It allows the user to dial up any combination of particles to collide (e, p, n, . . . but not nuclei), specify any reaction, and it generates very realistic event samples—lists of 4-vectors and particle ID codes for as many events as desired.

**Physical Insight**

Why do we need event generators at all? Because jets are defined by **jet-finding algorithms** that are implemented only as computer programs. The entire chain—from a hard parton scatter, through parton showering, to hadronization, to the observable spray of hadrons we call a "jet"—must be simulated numerically. PYTHIA does exactly this, starting from the perturbative (calculable) hard scatter and ending with the non-perturbative (modeled) conversion to hadrons.

## 1.2 Question 2: Hadronization Model in PYTHIA

**Question 2**

What is the name of the hadronization model used for the answer to 1) [i.e., used in PYTHIA]?

**Deliverable**

The name of the hadronization model implemented in PYTHIA.

**Background: Hadronization Models**

After the perturbative parton shower stage, the colored partons (quarks and gluons) must be converted into color-neutral hadrons. This **hadronization** step is intrinsically non-perturbative—we cannot calculate it from first principles using perturbation theory. Instead, we use phenomenological models.
There are two main hadronization models that have been used in event generators:

1. **String model (Lund string model):** Based on the idea that the color field between a quark and antiquark forms a **flux tube** (string) with a linear confining potential. As the $q\bar{q}$ pair separates, the string stretches and eventually breaks by producing new $q\bar{q}$ pairs, creating hadrons. Developed at Lund University in Sweden. Used in **PYTHIA** (and its predecessor JETSET).

2. **Cluster model:** After the parton shower, all gluons are non-perturbatively split into $q\bar{q}$ pairs. Color-singlet combinations naturally form low-mass "clusters" (due to the **preconfinement** property of parton showers). These clusters then decay isotropically into hadron pairs. Used in **HERWIG**.

**Source:** Frantz reading slides (p. 5), Webber talk (Section 3.2), Wong Ch. 7.

**Answer**

The **String Model** (also called the **Lund String Model**).
The Frantz slides state explicitly: PYTHIA "Uses 'String model' of hadrons/hadronization to provide framework but constants needed in this framework are 'tuned' empirically on other data."

**Physical Insight**

The string model is deeply connected to the physics of confinement. Because QCD is non-Abelian (gluons self-interact), the color electric field between a $q$ and $\bar{q}$ does not spread out like in QED. Instead, it collapses into a narrow **flux tube**—this has been confirmed by Lattice QCD calculations. The energy stored in this tube grows linearly with separation ($V \propto \kappa r$), which is exactly the behavior of a relativistic string with tension $\kappa \approx 1$ GeV/fm. When enough energy is stored, the string "breaks" by producing a new $q\bar{q}$ pair from the vacuum (the Schwinger mechanism), and the process repeats until only stable hadrons remain.

### 1.3 Question 3: Two Main Kinds of Jet Finding Algorithms

**Question 3**

What are the two main kinds of jet finding algorithms?

**Deliverable**

Two names: the two broad categories of jet-finding algorithms.

**Background: Why Do We Need Jet Algorithms?**

A jet is a collimated spray of hadrons resulting from the hadronization of a high-energy parton. But there is no unique, "obvious" way to look at a detector event and decide which particles belong to which jet, or even how many jets there are. We need a **jet-finding algorithm**: a well-defined mathematical procedure that takes a set of particle momenta as input and returns a set of jets as output.
A good jet algorithm must satisfy:

- **Infrared (IR) safety:** Adding a very soft particle should not change the jet configuration.
- **Collinear safety:** Splitting one particle into two collinear particles should not change the result.

These are essential because only IR- and collinear-safe observables can be meaningfully calculated in perturbative QCD.

**Source:** Vaidya jet slides (pp. 8–17 of reading PDF).

**Answer**

The two main kinds of jet finding algorithms are:

1. **Cone algorithms** — Define jets as all particles within a cone of fixed radius $R$ in $(\eta, \phi)$ space around a jet axis. The axis is found iteratively starting from a seed (usually the most energetic particle).
2. **Sequential recombination (clustering) algorithms** — Iteratively combine the "closest" pair of particles (according to a defined distance measure involving both angular separation and momentum) until some stopping criterion is reached. These work their way back through the parton splitting history.

### 1.4 Question 4: Which Algorithm Is No Longer Used?

**Question 4**

Which of the two answers in 3) is not actually used anymore (because it's not "infrared safe," e.g.)?

**Deliverable**

One of the two algorithm types from Q3 that is obsolete due to IR safety issues.

**Background: Infrared Safety Problem with Cone Algorithms**

The Vaidya slides (p. 14) show explicitly why the basic **iterative cone algorithm** is **not infrared safe**. The problem is this: adding a single soft particle between two hard particles can change whether the algorithm finds one jet or two. Because the algorithm uses seeds (the most energetic particles), a soft particle between two moderate-energy particles can shift the centroid of the cone enough to merge what were previously two separate jets into one, or vice versa.

The SIScone (Seedless Infrared-Safe Cone) algorithm was developed as an IR-safe version, but it scales computationally as $N^2 \ln N$, which is slow. In practice, at the LHC, clustering algorithms have completely replaced cone algorithms.

The slide header says it plainly: "Cone algorithms (**no one uses these anymore**)".

**Answer**

**Cone algorithms** are no longer used because the standard iterative cone algorithm is not infrared safe. Even the IR-safe variant (SIScone) is computationally expensive and has been superseded by sequential recombination algorithms at the LHC.

### 1.5 Question 5: Jet Algorithm Used at LHC

**Question 5**

Which jet finding algorithm is almost exclusively used by CMS and ATLAS at the LHC?

**Deliverable**

The specific name of the clustering algorithm used at ATLAS and CMS.

**Background: The Three Modern Clustering Algorithms**

The Vaidya slides (pp. 16–18) describe three "second-generation" sequential recombination algorithms, all defined by a single generalized distance measure:

$$d_{ij} = \min(k_{ti}^{2p}, k_{tj}^{2p}) \frac{\Delta_{ij}^2}{R^2}, \qquad d_{iB} = k_{ti}^{2p}$$

where $\Delta_{ij}^2 = (\eta_i - \eta_j)^2 + (\phi_i - \phi_j)^2$ is the angular separation, $k_{ti}$ is the transverse momentum of particle $i$, $R$ is the radius parameter, and the exponent $p$ distinguishes the three algorithms:

- $p = +1$: $k_t$ **algorithm** — clusters soft particles first (smallest $k_t$). Acts as the "inversion of the PYTHIA shower."
- $p = 0$: **Cambridge/Aachen (C/A) algorithm** — clusters closest particles first (purely angular). Acts as the "inversion of the HERWIG shower."
- $p = -1$: **anti-$k_t$ algorithm** — clusters *hard* particles first (largest $k_t$ seeds attract soft particles). Produces round, cone-like jets with stable boundaries.

All three are infrared and collinear safe.

**Answer**

The **anti-$k_t$ algorithm** (with $p = -1$) is almost exclusively used by ATLAS and CMS at the LHC.
The Vaidya slides state explicitly: anti-$k_t$ "Produces round jets" and is "Almost exclusively used by ATLAS and CMS."

**Why anti-$k_t$?**

The anti-$k_t$ algorithm is preferred because with $p = -1$, the distance measure $d_{ij} \propto \min(1/k_{ti}^2, 1/k_{tj}^2)$ is dominated by *hard* particles. This means hard particles "reach out" and grab soft particles nearby, producing jets with clean, circular boundaries in $(\eta, \phi)$ space—just like a cone algorithm would, but with the crucial advantage of being infrared and collinear safe. The jet area is stable and well-defined, which is essential for calibration and pileup subtraction at the LHC.

### 1.6 Question 6: The Variable $R$ and Its Formula

**Question 6**

What single letter capitalized variable is used to represent the angular size of jets and jet subcomponents, and what formula defines it? (in terms of $\phi$—angle difference—and what other quantity...)

**Deliverable**

1. The single capital letter variable name.
2. The formula for this variable.
3. The "other quantity" besides $\phi$.

**Background**

The Vaidya slides (p. 15, the annotation in blue text at the bottom) state:
"ON ALL THESE SLIDES THE VARIABLE **R** or Delta R (Rij means R difference between i and j) IS USED TO REPRESENT THE SIZES BETWEEN the PARTICLES/JETS...
THIS IS THE SAME R DEFINED BACK WHEN WE INTRODUCED RAPIDITY in terms

of PSEUDORAPIDITY (eta, the x-axis in the above plot)
dR = sqrt( dEtaˆ2 + dPhiˆ2)"
So the "other quantity" is **pseudorapidity** $\eta$ (or equivalently rapidity $y$ in the massless limit).

**Answer**

The variable is $\boldsymbol{R}$ (or $\Delta R$ for the separation between two objects $i$ and $j$). Its defining formula is:

$$\boxed{\Delta R = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}}$$

where:

- $\Delta\phi = \phi_i - \phi_j$ is the difference in azimuthal angle.
- $\Delta\eta = \eta_i - \eta_j$ is the difference in **pseudorapidity**.

This is the angular "distance" between two particles or jet subcomponents in the $(\eta, \phi)$ plane, which is the natural coordinate system for collider physics because differences in pseudorapidity are approximately Lorentz-invariant under boosts along the beam axis.

## 1.7 Question 7: Units of $R$ and Why

**Question 7**

*[This is the ONLY question NOT from the reading, but from the rapidity slides]:* What are the units of the answer to 6) [$R$], and why would/could the "other quantity" besides the $\phi$ angle mentioned in question 6 [i.e., pseudorapidity $\eta$] and thus $R$ itself be also measured in this unit? Be brief.

**Deliverable**

1. The units of $R$.
2. A brief explanation of why $\eta$ can be measured in the same units as an angle.

**Background: Pseudorapidity and Angles**

Recall the definition of pseudorapidity:

$$\eta = -\ln\left[\tan\left(\frac{\theta}{2}\right)\right]$$

where $\theta$ is the polar angle from the beam axis. This is a monotonic function of $\theta$, and there is a key identity:

$$\tan\left(\frac{\theta}{2}\right) = e^{-\eta}$$

Since $\eta$ is defined entirely in terms of the angle $\theta$ (no momentum or energy appears), it is a **purely angular quantity**. It is dimensionless, just like $\theta$ in radians.
Furthermore, $\phi$ (the azimuthal angle) is measured in **radians**. Since $\eta$ is also a dimensionless angular variable, $\Delta\eta$ and $\Delta\phi$ can be combined in the same quadrature sum, and $R$ inherits the same units.

**Answer**

The units of $R$ are **radians**.
The "other quantity" $\eta$ (pseudorapidity) can be measured in radians because $\eta$ is defined purely as a function of the polar angle $\theta$: $\eta = -\ln[\tan(\theta/2)]$. It is a dimensionless, purely geometric/angular quantity—no momentum or mass enters the definition. Since both $\Delta\eta$ and $\Delta\phi$ are dimensionless angular measures (both effectively in radians), the quantity $\Delta R = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}$ is also naturally measured in radians.

# 2 Part II: Fragmentation Functions (Questions 8–10)

## 2.1 Question 8: Argument of the Fragmentation Function $D()$

**Question 8**

What variable is the argument of the Fragmentation Function $D()$ and what is its physical significance—meaning what does its value (the argument variable) represent?

**Deliverable**

1. The variable name (letter).
2. Its physical meaning.

**Background: What Is a Fragmentation Function?**

A **Fragmentation Function (FF)** is the final-state analog of a Parton Distribution Function (PDF). While a PDF tells you the probability of finding a parton with momentum fraction $x$ inside an incoming hadron, a FF tells you the probability of finding a hadron carrying a certain fraction of the outgoing parton's momentum.
Formally, the quantity $D_1^{h/i}(z)\,dz$ gives the **number** of hadrons of type $h$ produced from a parton of type $i$ in the momentum fraction range $[z, z+dz]$.

**Sources:**

- Metz & Vossen review (arXiv:1607.02521), Section 1: "$D_1^{h/i}(z)\ldots$ describes the fragmentation of an unpolarized parton of type $i$ into an unpolarized hadron of type $h$, where the hadron carries the fraction $z$ of the parton momentum."
- Frantz reading slides on Fragmentation Functions (p. 24): "most universally is represented by the function $D(x)$ or $D(z)$ where $x$ or $z$ is supposed to represent the fraction of the OUTGOING parton's (q/gluon) momentum the jet's hadron ends up with."
- Webber talk, Section 2: The variable $x = 2E_h/E_{\text{cm}}$ in $e^+e^-$ annihilation.

**Answer**

The argument of the fragmentation function $D()$ is the variable $\boldsymbol{z}$ (or equivalently $x$ in some notations, especially for $e^+e^-$).
**Physical significance:** $z$ represents the **fraction of the outgoing parton's (quark or gluon) longitudinal momentum that the produced hadron carries**. That is, if the fragmenting parton has momentum $p_{\text{parton}}$, and the observed hadron has momentum $p_{\text{hadron}}$, then:

$$z = \frac{p_{\text{hadron}}}{p_{\text{parton}}} \quad \text{(longitudinal component along parton direction)}$$

with $0 < z \leq 1$.

**Important Distinction**

Do **not** confuse this $z$ (or $x$) with the Bjorken $x$ from DIS structure functions / PDFs! The PDF variable $x$ represents the fraction of the **incoming** hadron's momentum carried by the struck parton. The FF variable $z$ represents the fraction of the **outgoing** parton's momentum carried by the produced hadron. They are "mirror images" of each other in the factorization framework.

## 2.2 Question 9: Three Quantities in Factorization Convolution

**Question 9**

In one of the FF readings, an X with a circle around it represents INTEGRAL CONVOLUTION (not tensor product)...what three mathematical functions/quantities are generally convoluted in a factorization calculation? Which of them is not included for $e^+e^-$ annihilation factorization calculation and why?

**Deliverable**

1. Three functions/quantities in the general factorization formula.
2. Which one is absent for $e^+e^-$ annihilation and why.

**Background: QCD Factorization Theorems**

The Metz & Vossen review (p. 4 of their paper, p. 27 of reading PDF) gives the three key factorization formulas. The symbol $\otimes$ denotes integral convolution (integration over momentum fractions with appropriate kernels):
For the three main processes:

$$\sigma^{e^+e^-\to hX} = \hat{\sigma} \otimes FF \qquad \text{(SIA, Eq. 1)}$$

$$\sigma^{\ell N\to \ell hX} = \hat{\sigma} \otimes PDF \otimes FF \qquad \text{(SIDIS, Eq. 2)}$$

$$\sigma^{pp\to hX} = \hat{\sigma} \otimes PDF \otimes PDF \otimes FF \qquad \text{(pp, Eq. 3)}$$

Here:

- $\hat{\sigma}$ = the partonic cross section (perturbatively calculable hard-scattering kernel).
- $PDF$ = Parton Distribution Function (describes the structure of the incoming hadron).
- $FF$ = Fragmentation Function (describes hadron formation from outgoing parton).

**Answer**

The three mathematical functions/quantities generally convoluted in a factorization calculation are:

1. $\hat{\sigma}$ — the **partonic cross section** (the perturbatively calculable hard-scattering process).
2. **PDF** — the **Parton Distribution Function** (non-perturbative, describes initial-state hadron structure).

3. **FF** — the **Fragmentation Function** (non-perturbative, describes final-state hadronization).

**Which is absent for $e^+e^-$?** The **PDF** is not included for $e^+e^-$ annihilation.
**Why?** Because in $e^+e^-$ annihilation there are **no incoming hadrons**—the initial state consists of point-like leptons ($e^+$ and $e^-$), not composite hadrons. Since PDFs describe the partonic structure of incoming hadrons, they are simply not needed when the incoming particles are fundamental (structureless). The $e^+e^-$ process goes directly: $e^+e^- \to \gamma^*/Z \to q\bar{q} \to$ hadrons, and only the fragmentation of the outgoing quarks into hadrons requires a non-perturbative FF.

### 2.3 Question 10: Quark vs. Gluon Jets — DELPHI Results

**Question 10**

According to DELPHI measurements, for a bunch of jets all of the same momentum, which type—gluon or quark jets—will have particles with on average higher momentum?

**Deliverable**

One answer: quark or gluon.

**Background: Quark vs. Gluon Jet Differences**

The key difference between quark and gluon jets comes from the **color charges**: quarks carry color charge $C_F = 4/3$, while gluons carry $C_A = 3$. Since gluons have a larger color charge, they radiate more strongly, producing:

- **More particles** (higher multiplicity) in gluon jets.
- **Softer spectrum** (each individual particle carries less momentum) in gluon jets.
- **Broader** angular distribution in gluon jets.

Conversely, quark jets tend to have **fewer particles**, each carrying a **larger fraction** of the jet momentum, and the jet is more **collimated**.

**Source:** Webber talk, Section 5 (pp. 603–606): DELPHI selects gluon jets by anti-tagging heavy quark jets. The data clearly show: "Gluon jets have softer fragmentation than light quark jets, and higher multiplicity." The gluon/quark ratio of spectra in Fig. 12 shows the gluon spectrum is enhanced at low $x_E$ (soft particles) and suppressed at high $x_E$ (hard particles).

**Answer**

**Quark jets** will have particles with on average **higher momentum**.
For jets of the same total momentum, gluon jets produce more particles but each carries less momentum (softer fragmentation). Quark jets produce fewer particles, so each individual particle carries a larger share of the jet's total momentum. This is a direct consequence of the gluon's larger color charge ($C_A = 3$ vs. $C_F = 4/3$), which causes more intense radiation and more democratic sharing of energy among more fragments.

# 3 Part III: Schwinger Mechanism and String Model (Questions 11–18)

## 3.1 Optional Warmup: Eq. 5.2 and E&M

**Optional Warmup**

Is equation 5.2 ($\mathcal{E}A = q$) approximately the same as or not the same as the case of an infinite plane of charge in E&M?

**Background**

Wong Eq. 5.2 states: $\mathcal{E}A = q$, which relates the color electric field $\mathcal{E}$ inside the flux tube to the color charge $q$ via the cross-sectional area $A$ of the tube. This is a statement of **Gauss's law**: the electric flux through a closed surface equals the enclosed charge.
For an **infinite plane of charge** in classical E&M, Gauss's law gives $E \cdot A = \sigma A/\epsilon_0$, resulting in a *uniform* electric field $E = \sigma/(2\epsilon_0)$ that is constant and independent of distance. Similarly, inside the flux tube, $\mathcal{E}$ is constant—the field does not spread out because the tube has a fixed cross section $A$.

**Answer**

**Yes**, Eq. 5.2 is essentially the same as (i.e., it is an application of) Gauss's law, just as in the case of an infinite plane of charge in E&M. In both cases, the electric field is **uniform** (constant magnitude, independent of position). For the plane, the field is uniform because of the infinite planar symmetry. For the flux tube, the color electric field is uniform because the non-Abelian nature of QCD confines the field lines into a tube of fixed cross-sectional area $A$, rather than allowing them to spread out as $1/r^2$.

## 3.2 Question 11: Order of Magnitude of String Tension $\kappa$

**Question 11**

What is the order of magnitude of the string tension constant $\kappa$ on page 63 of Wong?

**Deliverable**

The numerical value and units of $\kappa$.

**Background**

Wong p. 63 (p. 43–44 of reading PDF) introduces the string tension $\kappa$ through Eq. 5.1:

$$\frac{1}{2}\mathcal{E}^2 \cdot A = \kappa$$

The string tension represents the energy stored per unit length in the color flux tube. It can be estimated from the Regge slope parameter $\alpha'$ and from hadron spectroscopy (the linear potential $V(r) = \kappa r$ fitted to charmonium and bottomonium spectra).
Wong states: "the string tension $\kappa$, which has been estimated from the Regge slope parameter to be about 1 GeV/fm."

**Answer**

$$\boxed{\kappa \approx 1 \text{ GeV/fm}}$$

This means that separating a quark from an antiquark by 1 femtometer requires approximately 1 GeV of energy stored in the color flux tube. For context, this is an enormous energy density—pulling two quarks apart by just $\sim$1 fm stores roughly the rest-mass energy of a proton.

### 3.3 Question 12: Difference Between Eq. 5.4 and 5.5

**Question 12**

What is the difference between Eq.'s 5.4 and 5.5, and what physical phenomenon causes it?

**Deliverable**

1. The mathematical difference between the two equations.
2. The physical phenomenon responsible.

**Background**

Wong p. 65 (p. 44 of reading PDF) considers the potential experienced by a produced quark $q_1$ inside the flux tube between the original $q_0$ and $\bar{q}_0$:
**Eq. 5.4:** If a **single** quark $q_1$ is produced (unphysical—would violate charge conservation):

$$V(z) = -q\mathcal{E}z = -2\kappa z \quad \text{(single } q_1 \text{ produced, unphysical)}$$

The force on $q_1$ is $2\kappa$: a "push" of $\kappa$ from $q_0$ at $z = 0$ and a "pull" of $\kappa$ from $\bar{q}_0$ at $z = L$.
**Eq. 5.5:** In reality, the produced quark $q_1$ is **always accompanied** by a produced antiquark $\bar{q}_1$. The $\bar{q}_1$ **screens** the force from $q_0$ on $q_1$—the newly produced $\bar{q}_1$ cancels the repulsive "push" from $q_0$:

$$V(z) = -\kappa z \quad \text{(additional antiquark } \bar{q}_1 \text{ interaction included)}$$

The net force on $q_1$ is now only $\kappa$ (just the pull from $\bar{q}_0$), reduced by a factor of 2 compared to 5.4.

**Answer**

**Difference:** Eq. 5.4 gives $V(z) = -2\kappa z$ (force $= 2\kappa$), while Eq. 5.5 gives $V(z) = -\kappa z$ (force $= \kappa$). The potential is reduced by a factor of 2.
**Physical cause: Color charge screening.** A single quark is never produced alone—charge conservation requires that the production of quark $q_1$ is always accompanied by an antiquark $\bar{q}_1$. This newly produced $\bar{q}_1$ sits between $q_0$ and $q_1$ and **screens** (cancels) the repulsive force that $q_0$ would otherwise exert on $q_1$. The result is that the effective potential experienced by $q_1$ is halved.

### 3.4 Question 13: Diagram for Each Value of WHAT?

**Question 13**

Wong says that there is a diagram like in Figure 7.1a/b for each value of WHAT that the exchanged or virtually produced (aka s-channel) particle possesses.

**Deliverable**

The physical quantity (one word or phrase) that labels different diagrams of the type in Fig. 7.1.

**Background**

Wong p. 107–108 (pp. 48–49 of reading PDF) discusses the dual resonance model of elastic meson-meson scattering. The s-channel diagram (Fig. 7.1a) shows two incoming mesons forming an intermediate particle $p_J$, which then decays. The t-channel diagram (Fig. 7.1b) shows the exchange of a particle $p_J$.
Wong writes: "The intermediate particle $p_J$ in Fig. 8.1a or the exchanged particle $p_J$ in Fig. 8.1b can possess a spin. There are problems when this intermediate particle or exchange particle possesses a large spin $J$ ... There is a diagram of the type (7.1a) for each value of the spin $J$ of the intermediate particle."

**Answer**

There is a diagram for each value of the **spin** $J$ (angular momentum) that the exchanged or intermediate particle possesses.

### 3.5 Question 14: Regge Trajectory

**Question 14**

A Regge trajectory is a _____ (what kind of "graphical" relationship) between the answer to 13) [$J$] and _____ (what?).

**Deliverable**

Two blanks: (1) the type of graphical relationship, (2) the other variable.

**Background**

Wong p. 109 (p. 49 of reading PDF) and Fig. 7.2 show the Regge trajectory relation. Hadron resonances with the same internal quantum numbers but different spins fall onto straight lines when one plots $J$ vs. $s = M^2$ (mass squared). The relation is:

$$J(M) = \alpha(0) + \alpha' M^2$$

where $\alpha(0)$ is the Regge intercept and $\alpha' \approx 1$ GeV$^{-2}$ is the Regge slope. Fig. 7.2 shows this for baryons: plotting $J$ on the vertical axis against $s = M^2$ on the horizontal axis gives approximately straight lines.

**Answer**

A Regge trajectory is a **linear (straight line)** relationship between the **spin** $J$ and the **mass squared** $M^2$ (equivalently $s = M^2$) of the hadron:

$$J = \alpha(0) + \alpha' M^2$$

The Regge slope $\alpha' \approx 1$ GeV$^{-2}$ is approximately universal across meson and baryon families.

### 3.6 Question 15: Experimental Data the String Model Cannot Explain

**Question 15**

Name one kind of experimental data Wong says the string model cannot explain well.

**Answer**

Wong (p. 111) lists several shortcomings of the string theory of strong interactions. Notably, the string model has difficulties explaining:
**Hard processes with large transverse momentum transfers observed in deep inelastic electron scattering experiments**—i.e., the partonic structure revealed by DIS. The string model predicts a massless vector particle $\rho$ and a massless tensor particle, but cannot properly account for the pointlike parton behavior at short distances that led to the concept of partons in DIS. Wong states the model "has difficulties in explaining many hard processes with large transverse momentum transfers."
(Any of the difficulties Wong mentions is acceptable: failure with DIS/large-$p_T$ processes, prediction of incorrect massless particles, etc.)

### 3.7 Question 16: Figure 7.4(a) — Slopes and Rest/Motion

**Question 16**

In Figure 7.4(a), what are the values of the slopes of the line segments, and is the meson the diagram is supposed to represent at rest or moving?

**Deliverable**

1. The numerical values of the slopes of the quark and antiquark trajectories.
2. Whether the meson is at rest or moving.

**Background**

Wong §7.2, Fig. 7.4(a) (p. 113, reading PDF p. 51) shows the space-time trajectories of a quark (solid line) and antiquark (dashed line) for a **stable meson at rest** (the yo-yo state).
From Eq. 7.7: the velocity of each particle is $\dot{x}_i = \text{sign}(p_i)$, i.e., they travel at the **speed of light** (the quarks are treated as massless). So the slopes of the trajectories in the $(x, t)$ plot—where slope $= dt/dx = 1/v$—correspond to particles moving at $v = \pm c$.
The quark starts moving in the $+x$ direction at $v = +c$ (slope $= +1$ in the $t$ vs. $x$ plot), then

turns around and moves at $v = -c$ (slope $= -1$). The antiquark does the mirror image.

**Answer**

1. The slopes of the line segments are $\mathbf{+1}$ and $\mathbf{-1}$ (in the $t$ vs. $x$ diagram), corresponding to the quark and antiquark traveling at the **speed of light** ($v = \pm c$), since they are treated as massless particles.

2. The meson is **at rest** (it is in its center-of-mass frame; this is the "yo-yo state at rest"). Fig. 7.4(a) is explicitly labeled for the yo-yo at rest. Fig. 7.5(b) shows the yo-yo *in motion*.

### 3.8 Question 17: The Yo-Yo Model

**Question 17**

Describe what oscillation of the string which the "yo-yo model" describes, and what quantity/property of the meson the string represents. Can the energy associated with this oscillation be related to?

**Deliverable**

1. Description of the yo-yo oscillation.

2. The meson property the string represents.

3. What the oscillation energy relates to.

**Background**

Wong §7.2 (pp. 112–115, reading PDF pp. 51–53) introduces the yo-yo model. A meson at rest is modeled as a massless quark and antiquark connected by a string (flux tube) with tension $\kappa$. The quark and antiquark:

1. Start at the same point.

2. Fly apart at the speed of light, stretching the string.

3. The string tension decelerates them (their momenta decrease linearly with time).

4. They stop, reverse direction, and fly back together.

5. They meet, pass through each other, and the cycle repeats.

This is exactly like a yo-yo: periodic expansion and contraction. The period is $T = 4|p_q(0)|/\kappa$ (Eq. 7.13).

**Answer**

1. **The oscillation:** The yo-yo model describes the periodic **expansion and contraction** of a quark-antiquark string. The $q$ and $\bar{q}$ fly apart at the speed of light, stretching the string;

the string tension $\kappa$ decelerates them until they stop; then they reverse direction and fly back together; they meet, pass through each other, and the cycle repeats. This oscillatory motion is the origin of the name "yo-yo state."

2. **Meson property:** The string represents the **confining color flux tube** between the quark and antiquark. Its tension $\kappa$ represents the linear confining potential of QCD.

3. **Energy relation:** The total energy of the yo-yo system is $H = 2|p_q(0)|$ (Eq. 7.15), which is identified as the **rest mass** $m$ of the stable meson (since $H = \sqrt{s}$ of the $q\bar{q}$ system at rest). Furthermore, the angular momentum of the string's rotation (a rotating string) can be related to the spin $J$ of the meson, yielding the Regge trajectory $J = \alpha' M^2$, connecting the oscillation energy (mass) to the angular momentum (spin).

### 3.9 Question 18: Geometric Curve for Particle Production Points (Lund Model)

**Question 18**

What kind of geometric curve (a conic section) does the Lund model propose that all particle production points in spacetime fall along approximately? Which part of the graph/trajectory is traced out on figure 7.6—upper or lower part?

**Deliverable**

1. The name of the conic section.

2. Upper or lower part of Fig. 7.6.

**Background**

Wong §7.4 (p. 120, reading PDF p. 54) discusses the symmetric Lund model. The key assumption is: "the vertices at which the quark and antiquark pairs are produced lie approximately on a curve of constant proper time." The proper time $\tau$ is related to $t$ and $x$ by:

$$\sqrt{\Gamma} = \kappa\tau \quad \text{where} \quad \Gamma = p_+ p_- = \kappa^2 uv = \kappa^2(t^2 - x^2)$$

So the condition of constant proper time $\tau = \tau_0$ gives:

$$t^2 - x^2 = \text{const}$$

This is the equation of a **hyperbola** in spacetime (a conic section).
Looking at Fig. 7.6 (p. 54 of reading PDF), the production vertices $V_0, V_1, V_2, \ldots$ lie along the **lower part** of the diagram (near the origin in spacetime, at early times).

**Answer**

1. The particle production points fall approximately along a **hyperbola** (a conic section defined by $t^2 - x^2 = \text{const}$, i.e., a curve of constant proper time $\tau$).

2. The production vertices are traced out on the **lower part** of the graph/trajectory in Fig. 7.6 (near the spacetime origin, where the string fragmentation vertices $V_i$ appear).

# 4 Part IV: Dual Parton Model / Pomeron (Questions 19–23)

## 4.1 Question 19: Hadron Produced in Diffractive Events

**Question 19**

What kind of hadron is likely to be produced in an inelastic (but "pseudo-elastic," i.e., proton intact) diffractive event/interaction?

**Answer**

**Vector mesons** (such as $\rho$, $\omega$, $\phi$, $J/\psi$).
This is because in diffractive events, the exchanged Pomeron couples through a photon-like mechanism (the photon in the process is a vector particle with $J^{PC} = 1^{--}$), so the dominant production is of hadrons with the same quantum numbers—i.e., **vector mesons**. This principle is known as **Vector Meson Dominance (VMD)**.
From the Frantz slides (p. 58–59 of reading PDF): "because the photon in the process is a vector particle, the dominant production will be of VECTOR mesons. This is known as Vector Meson Dominance VMD."

## 4.2 Question 20: Color Dipole Model

**Question 20**

What perturbative model describes the interaction of a virtual photon splitting into a color/anticolor charge pair which interacts with the hadron? Hint: the slides say Color Transparency, a somewhat well known but also perhaps somewhat maligned theory sometimes used to describe lots of JLab data, is an offshoot of this model.

**Answer**

The **Color Dipole Model**.
From the Frantz slides (p. 60 of reading PDF): "Closely related to VMD is a family of perturbative theories known as the **Color Dipole Model**. In this model... the photon... interacts with the incoming proton as a $q\bar{q}$ pair 'fluctuation'..." The slides note that Color Transparency is "one of its off-shoots" (associated with Kopeliovich).

## 4.3 Question 21: Veneziano Amplitude Symmetry

**Question 21**

What two variables is the Veneziano scattering amplitude expression (Eq. 8.1) invariant to the exchange of, reminiscent of the exchanges/permutations of different Feynman diagram results for various kinds of "crossing" relations?

**Answer**

The Veneziano amplitude $A(s,t)$ (Eq. 8.1) is invariant under the exchange of the Mandelstam

variables $\boldsymbol{s}$ and $\boldsymbol{t}$:

$$A(s,t) = A(t,s)$$

This "duality" property means the s-channel and t-channel descriptions give the *same* amplitude—what would be two separate Feynman diagrams in perturbation theory are unified into a single expression. This is the origin of the name "dual resonance model."
Wong (p. 134): "The Veneziano amplitude (8.1) is invariant with respect to the interchange of $s$ and $t$."
Furthermore, the full crossing-symmetric amplitude $A_4 = A(s,t) + A(t,u) + A(u,s)$ (Eq. 8.4) is invariant under exchange of all three Mandelstam variables $s$, $t$, and $u$—but the question specifically asks about Eq. 8.1, which has $s \leftrightarrow t$ symmetry.

### 4.4 Question 22: Total Cross Section vs. CM Energy

**Question 22**

What kind of relationship does the reading in Wong's D.P.M. section ascribe between the total cross section and the scattering center of mass energy?

**Answer**

The total cross section is related to the center-of-mass energy $\sqrt{s}$ by a **power-law** relationship. Specifically, from Wong Eq. 8.9:

$$\boxed{\sigma_{\text{tot}} \propto s^{\alpha_0 - 1}}$$

where $\alpha_0$ is the Regge intercept of the leading trajectory.
For the Pomeron (with $\alpha_0 \approx 1$), this gives $\sigma_{\text{tot}} \approx$ const (approximately energy-independent, or very slowly rising). For other Reggeon exchanges with $\alpha_0 < 1$, the cross section *falls* with energy.
The key point is that the total cross section approaches a **nearly constant value** at very high energies (when Pomeron exchange dominates), which is one of the defining experimental features that motivated the Pomeron concept.

### 4.5 Question 23: Pomeron Quantum Numbers

**Question 23**

Wong says the Pomeron has the quantum numbers of the _____ (what)?

**Answer**

The Pomeron has the quantum numbers of the **vacuum**.
Wong (p. 141) discusses properties of hadronic reactions at high energies: "(i) inelastic processes with no quantum number flow appear to dominate" and "Property (i) suggests that the exchange of an object with the quantum numbers of the vacuum can be a useful concept for forward reactions." The Pomeron is precisely this object: a Regge trajectory with $\alpha_0 \approx 1$ and vacuum quantum numbers (zero charge, zero strangeness, zero baryon number, isospin $I = 0$, and $C = P = +1$).

# 5 Part V: MIT Bag Model and QGP (Questions 24–28)

## 5.1 Question 24: Important Variables in the Bag Model

**Question 24**

List some variables/quantities whose values are important in the Bag Model.

**Answer**

Key variables/quantities in the MIT Bag Model include:

- $\boldsymbol{B}$ — the **bag pressure constant** (energy density difference between the vacuum outside and inside the bag). This is the single free parameter of the model.
- $\boldsymbol{R}$ — the **radius of the bag** (the confinement radius).
- $\boldsymbol{N}$ (or $N_q$) — the **number of quarks** inside the bag (3 for baryons, 2 for mesons).
- $\boldsymbol{x_n}$ — the **eigenvalue** from the boundary condition of the Dirac equation on the bag surface ($x_1 = 2.04$ for the lowest mode).
- $E_{\text{kin}}(R) = N_q \cdot x_n/R$ — the **kinetic energy** of the confined quarks.
- $E_{\text{pot}}(R) = \frac{4}{3}\pi R^3 B$ — the **potential energy** from the bag pressure.
- $\boldsymbol{p^0} = x_n/R = 2.04/R$ — the quark **momentum eigenvalue** inside the bag.

## 5.2 Question 25: What Does $B$ Represent?

**Question 25**

What does the constant $B$ represent in the Bag Model?

**Background**

In the MIT Bag Model, quarks are treated as massless particles confined inside a spherical "bag" of radius $R$. Inside the bag, quarks move quasi-freely (asymptotic freedom at short distances). Confinement is enforced by a **boundary condition** at the bag surface that prevents quarks from escaping.
The physical picture: the "vacuum" outside the bag has a higher energy density than the region inside (where the perturbative vacuum with free quarks exists). This energy density difference creates an inward pressure that confines the quarks.

**Sources:** Bag model slides (pp. 69–73 of reading PDF), Wong §9.2 (pp. 159–161).

**Answer**

$B$ represents the **bag pressure** (or equivalently, the **bag constant**)—it is the difference in energy density between the non-perturbative QCD vacuum outside the bag and the perturbative vacuum inside the bag.
Physically, $B$ is the **inward pressure** exerted by the external vacuum on the bag surface,

which is what confines the quarks inside. The potential energy stored in a bag of radius $R$ is $E_{\text{pot}} = \frac{4}{3}\pi R^3 B$, representing the "cost" of maintaining a region of perturbative vacuum (the bag interior) against the confining pressure of the non-perturbative vacuum.
From the bag model slides (p. 72): "B—bag constant that reflects the bag pressure."

### 5.3 Question 26: How the $1/4$ Power Arises

**Question 26**

How does the $\frac{1}{4}$ power arise in relation to the expressions for $B$? (i.e., in the derivation of those equations, relating what to what, or finding what…what part of those derivations)

**Background: The Bag Model Energy Minimization**

The total energy of $N$ quarks in a bag of radius $R$ is (Wong p. 161, slides p. 72–73):

$$E(R) = E_{\text{kin}} + E_{\text{pot}} = \frac{N \cdot x_n}{R} + \frac{4}{3}\pi R^3 B$$

The first term (kinetic) decreases with $R$ (quarks want to spread out), and the second term (bag pressure potential) increases with $R$ (the bag resists expansion). The equilibrium radius $R_n$ is found by **minimizing** $E(R)$:

$$\frac{dE}{dR} = -\frac{N x_n}{R^2} + 4\pi R^2 B = 0$$

Solving for $R$:

$$4\pi R^4 B = N x_n \quad \Longrightarrow \quad R_n = \left(\frac{N_q x_n}{4\pi B}\right)^{1/4}$$

This is where the $1/4$ **power** arises.

**Answer**

The $\frac{1}{4}$ power arises from **minimizing the total energy** $E(R)$ of the bag with respect to the bag radius $R$.
The total energy has two competing terms: $E_{\text{kin}} \propto 1/R$ (which decreases with $R$) and $E_{\text{pot}} \propto R^3$ (which increases with $R$). Setting $dE/dR = 0$ yields an equation of the form $R^4 \propto$ (constants), so solving for $R$ requires taking the **fourth root**—hence the $1/4$ power:

$$R_n = \left(\frac{N_q x_n}{4\pi B}\right)^{1/4}$$

Similarly, substituting back to get $B^{1/4}$:

$$B^{1/4} = \left(\frac{2.04N}{4\pi}\right)^{1/4} \frac{1}{R}$$

The $1/4$ power thus comes from balancing a $1/R$ kinetic term against an $R^3$ potential term: $1/R^2 = R^2 \implies R^4 = \text{const}$.

## 5.4 Question 27: Two Physical Quantities Derived from the Bag Model

**Question 27**

Name two physical quantities (any of the answers to 24 are OK) that are derived in the reading selections/slides using the Bag Model, and what are their values?

**Answer**

Two quantities derived from the Bag Model:

1. **The bag pressure constant $B^{1/4}$:** By fitting the model to the nucleon mass ($M_N = 938$ MeV) with $N_q = 3$ quarks and a confinement radius $R \approx 0.8$ fm (from Wong p. 161):

$$\boxed{B^{1/4} \approx 206 \text{ MeV}}$$

   (The value falls in the range 145–235 MeV depending on the fitting procedure.)

2. **The critical temperature for QGP formation $T_c$:** Using the bag model estimate (Wong Eq. 9.8, p. 164):

$$T_c = \left(\frac{90}{37\pi^2}\right)^{1/4} B^{1/4}$$

   For $B^{1/4} = 206$ MeV:

$$\boxed{T_c \sim 144 \text{ MeV}}$$

Other acceptable answers include: the nucleon radius ($R \approx 0.8$ fm), the quark momentum eigenvalue ($p^0 = 2.04/R$), or the nucleon mass itself (used as input to fix $B$).

## 5.5 Question 28: Critical Temperature for QGP Formation

**Question 28**

What is the critical temperature for QGP (quark-gluon plasma) formation Wong gives from Bag model estimates?

**Deliverable**

The numerical value of $T_c$ from the bag model.

**Background: QGP Phase Transition in the Bag Model**

Wong §9.3.1 (pp. 162–164) considers a quark-gluon system at high temperature $T$ in thermal equilibrium. The total pressure of the ideal quark-gluon plasma (treating quarks and gluons as massless) is (Eq. 9.6):

$$P = g_{\text{total}} \frac{\pi^2}{90} T^4$$

where the total degeneracy factor is (Eq. 9.3):

$$g_{\text{total}} = \left[ g_g + \frac{7}{8}(g_q + g_{\bar{q}}) \right] = \left[ 16 + \frac{7}{8}(12 + 12) \right] = 37$$

Here: $g_g = 8 \times 2 = 16$ (8 gluon colors $\times$ 2 polarizations), $g_q = g_{\bar{q}} = N_c \times N_s \times N_f = 3 \times 2 \times 2 = 12$ (for 2 flavors).
The QGP phase transition occurs when the outward pressure of the quark-gluon plasma equals the inward bag pressure $B$:

$$P = B \quad \Longrightarrow \quad 37 \frac{\pi^2}{90} T_c^4 = B$$

**Answer**

$$T_c = \left( \frac{90}{37\pi^2} \right)^{1/4} B^{1/4}$$

For $B^{1/4} = 206$ MeV (from the nucleon mass fit):

$$\boxed{T_c \approx 144 \text{ MeV}}$$

Wong notes (p. 164): "Thus, for $B^{1/4} = 206$ MeV as estimated in Section 9.2, we have $T_c \sim 144$ MeV."

**Comparison with experiment:** The reading notes that the real transition temperature found at RHIC was about **170–180 MeV**, somewhat higher than this simple bag model estimate. More refined lattice QCD calculations give $T_c \approx 155$–$170$ MeV depending on the number of quark flavors and other details. The bag model gives the right order of magnitude but underestimates $T_c$ because it is a highly simplified model (massless quarks, no interactions, sharp bag boundary, etc.).

**The Big Picture: Bag Model → QGP**

The bag model provides an elegant, intuitive picture of the QGP transition: if you heat hadronic matter enough (or compress it enough), the pressure of the quarks and gluons inside individual bags eventually exceeds the confining bag pressure $B$. At that point, the bags "pop"—individual hadron bags merge into a single large volume of deconfined quarks and gluons. This is the quark-gluon plasma. The critical temperature where this happens is set by the balance $P_{\text{QGP}} = B$.

# 6 Summary Table

| Q | **Short Answer** |
|---|---|
| 1 | PYTHIA |
| 2 | String Model (Lund String Model) |
| 3 | Cone algorithms and Sequential recombination (clustering) algorithms |
| 4 | Cone algorithms (not IR safe) |
| 5 | Anti-$k_t$ algorithm |
| 6 | $R$; $\Delta R = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}$; other quantity = pseudorapidity $\eta$ |
| 7 | Radians; $\eta$ is a purely angular quantity defined via $\theta$ |
| 8 | $z$ (or $x$); fraction of outgoing parton's momentum carried by hadron |
| 9 | $\hat{\sigma}$, PDF, FF; PDF absent for $e^+e^-$ (no incoming hadrons) |
| 10 | Quark jets (higher avg. particle momentum; gluon jets are softer) |
| 11 | $\kappa \approx 1$ GeV/fm |
| 12 | Eq. 5.5 has half the potential of 5.4; caused by color screening from produced $\bar{q}_1$ |
| 13 | Spin $J$ |
| 14 | Linear (straight line) relationship between $J$ and $M^2$ |
| 15 | Hard processes / large transverse momentum transfers / DIS data |
| 16 | Slopes $= \pm 1$ (speed of light); meson at rest |
| 17 | Expansion/contraction of $q\bar{q}$ string; energy $\rightarrow$ meson rest mass |
| 18 | Hyperbola ($t^2 - x^2 = \text{const}$); lower part of Fig. 7.6 |
| 19 | Vector mesons (VMD) |
| 20 | Color Dipole Model |
| 21 | $s$ and $t$ (Mandelstam variables) |
| 22 | Power law: $\sigma_{\text{tot}} \propto s^{\alpha_0 - 1}$ ($\approx$ const for Pomeron with $\alpha_0 \approx 1$) |
| 23 | The vacuum |
| 24 | $B$, $R$, $N_q$, $x_n$, $E_{\text{kin}}$, $E_{\text{pot}}$, $p^0$ |
| 25 | Bag pressure (vacuum energy density difference confining quarks) |
| 26 | Minimizing $E(R)$: balancing $1/R$ vs. $R^3$ gives $R^4 \propto$ const $\rightarrow 1/4$ power |
| 27 | $B^{1/4} \approx 206$ MeV; $T_c \approx 144$ MeV (or $R \approx 0.8$ fm, etc.) |
| 28 | $T_c \approx 144$ MeV (RHIC found $\sim$170–180 MeV) |